PCCP

Physical Chemistry Chemical Physics

Accepted Manuscript

This article can be cited before page numbers have been issued, to do this please use: S. Ntim and M. Sulpizi, *Phys. Chem. Chem. Phys.*, 2020, DOI: 10.1039/D0CP00409J.

View Article Online
DOI: 10.1039/D0CP00409J

Journal Name

ARTICLE TYPE

Cite this: DOI: 00.0000/xxxxxxxxxx

Role of Image Charges in Ionic Liquid Confined between Metallic Interfaces

Samuel Ntim and Marialore Sulpizi*

Received Date
Accepted Date

DOI: 00.0000/xxxxxxxxxx

The peculiar properties of ionic liquids in confinement have not only become essential for energy storage, catalysis and tribology, but still pose fundamental questions. Recently, an anomalous liquid-solid phase transition has been observed in atomic force microscopy experiments for 1-butyl-3-methylimidazolium tetrafluoroborate ([BMIM][BF_4]), the transition being more pronounced for metallic surfaces. Image charges have been suggested as the key element driving the anomalous freezing. Using atomistic molecular dynamics simulations, we investigate the impact of image charges on structure, dynamics and thermodynamics of [BMIM][BF_4] confined between gold electrodes. Our results not only unveil a minor role played by the metal polarisation, but also provide a novel description of the interfacial layer. Although no diffuse layer can be defined in terms of the electrostatic potential, long range effects are clearly visible in the dynamical properties up to 10 nanometers away from the surface, and are expected to influence viscous forces in the experiments.

1 Introduction

Since the nineties room temperature ionic liquids (RTILs) have attracted constant interest from both the fundamental point of view as models for concentrated electrolytes, as well as for their applications. In particular the structure of RTILs in confinement has raised interest for its critical role in energy storage and transformation in supercapacitors,[1,2] actuators,[3,4] batteries,[5–7] and solar cells.[8] The presence of an interface with a solid surface modifies the liquid properties producing a layered structure, whose extent strongly depends on the liquid composition as well as on the nature and structure of the surface.[9] The presence of layers for several types of surfaces has been observed both in experiments, including X-ray reflectivity,[10] Atomic Force Microscope (AFM)[11,12] and Surface Force Balance (SFB),[13–15] as well as in simulations.[16] On the other hand, the presence of a diffuse layer, in analogy to what is found in dilute electrolytes, is still a debated issue in the community.[17–27] Therefore the detailed characterisation of the interfacial layer is still far from being clear and unanimous.

Additionally, due to the dominant role of electrostatic forces, one may anticipate that the metallic nature of the confining surfaces should affect the static and dynamic properties of confined RTILs. Such a relationship has been recently explored using tuning-fork-based atomic force microscope nanorheological experiments.[23] These revealed a dramatic change of the RTIL towards a solid-like phase below a threshold thickness related to the metallic nature of the confining materials, with more metallic surfaces facilitating capillary freezing. Siria and coworkers also proposed a simple explanation of the observed phenomena in terms of image charges.[23] The simple idea highlighted in Ref. 23 is that if one considered a semi-infinite ionic crystal at the interface with a perfect metal, the network of image charges would build a crystal structure with an almost perfect symmetry with respect to the real half-lattice. Accordingly, one would expect the electrostatic contribution to the surface free energy to nearly vanish, as the system behaves as a single bulk lattice. In real systems further complications arise from the shape of the ions and their relative size. Moreover, as recently pointed out by Lhermerout and Perkin[26] in the SFB experiments it is difficult to separate the electrostatic and the viscous contributions to the measured forces. Ionic liquids may have dynamic viscosity even 100 times greater than that of water (as e.g. in the case of 1-butyl-3-methylimidazolium cation ([BMIM]$^+$) and tetrafluoroborate anion ([BF_4]$^-$) ([BMIM][BF_4]), and consequently may exhibit viscous forces of the same order of magnitude as the reported electrostatic forces at the approach velocities typically employed in experiments.

Our aim here is to investigate the role of the image charges by performing atomistic molecular dynamics simulations of ionic liquid in confinement. We consider simple equilibrium conditions in the absence of viscous forces. We choose the [BF_4]$^-$

Institut für Physik, Johannes Gutenberg Universität, Staudingerweg 7, 55128-Mainz, Germany; E-mail: sulpizi@uni-mainz.de

† Electronic Supplementary Information (ESI) available: [details of any supplementary information available should be included here]. See DOI: 00.0000/00000000.

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DOI: 10.1039/D0CP00409J

Fig. 1 Typical simulation box of 1200 pairs of [BMIM][BF_4] confined in non-polarisable gold. Gold atoms are replaced with polarisable gold "molecules" for simulations with polarisable gold.

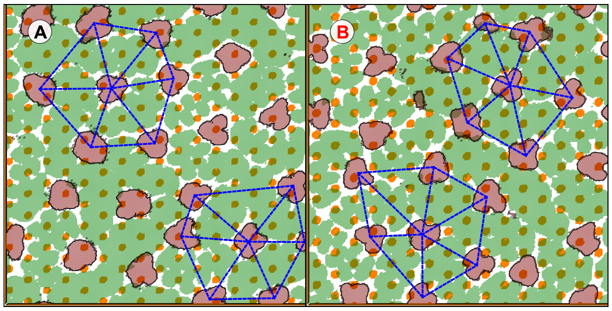

Fig. 2 Density map of adsorbed ions on (A) non-polarisable gold surface and (B) polarisable surface. Yellow dots represent the first layer of gold atoms, the green map corresponds to the cation while the brown map corresponds to the anion. The blue dashed lines are to emphasise the defective hexagonal arrangement of the anions.

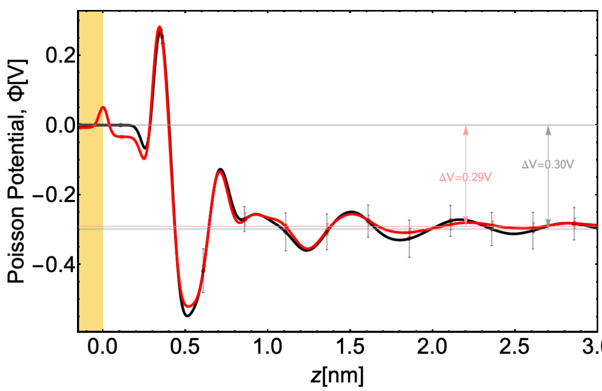

Fig. 3 Potential drop in liquid on non-polarisable gold surface (black) and on polarisable gold surface (red).

[BMIM][BF_4]) system (as in Ref. 23) confined between two Au(111) surfaces. To include the image charge effects we use the method which we have recently developed and which permits to describe the metal polarisation in a consistent way without additional computational costs.[28] Alternatively, a fixed potential approach could also be used[29] or the Electrostatic Layer Correction with Image Charges (ELCIC).[30] A snapshot of our model system is reported in Figure 1. We analyse structural, energetic and dynamical properties of the RTIL/gold interfaces with and without image charges. In particular, we aim to address the role of image charges on the extent of the interfacial layering of the RTIL, to quantify the influence of image charges on the interfacial free energy and finally to investigate the influence of image charges on dynamical properties, such e.g. the diffusion coefficients.

2 Results and discussion

As already well established, the presence of an interface induces a strong layering in the confined RTIL.[9,13,16,32,33] This is also confirmed in our simulations and shown by the total mass density profile reported in the inset of Figure **??** where the layering is observed to extend for about 2 nm from the surface. Interestingly

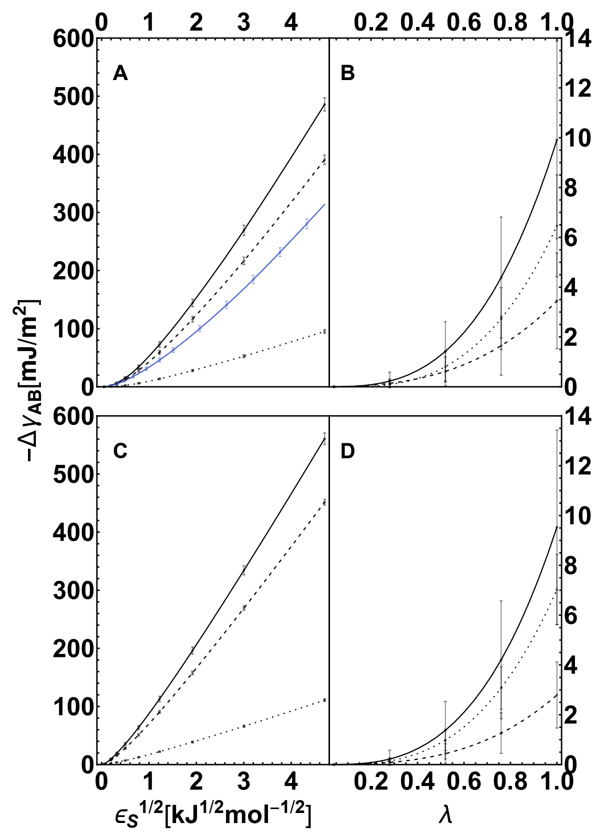

Fig. 4 Work of adhesion due to (A) Lennard-Jones gold-ionic liquid interactions and (B) electrostatic gold-ionic liquid interactions. Continuous black curves are the sums of their corresponding dashed (BMIM) and dotted (BF_4) curves. For sake of comparison, the blue curve in (A) represents the work of adhesion for the (non-polarisable) gold-SPC/E water interface and is very close to that reported in Ref. 31. (C) and (D) are works of adhesion due to Lennard-Jones and electrostatic interactions respectively but at 150 K.

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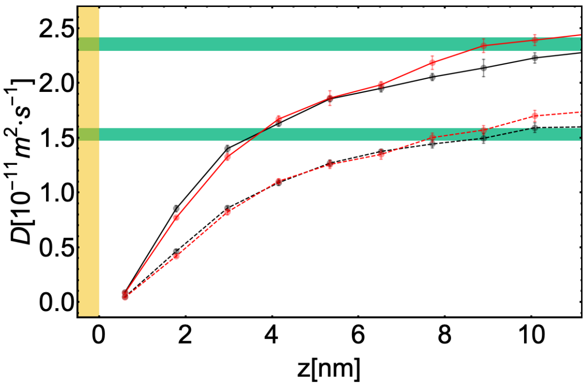

Fig. 5 Diffusion coefficient of the ions, across the interface. For each trajectory, we divided the symmetrised region between the gold slabs into ten regions and calculated the 2-dimensional diffusion coefficient in the plane of the gold surface (x-y plane) every 2.5 ns for 25 ns and averaged. Solid lines represent cation diffusion coefficient while dashed lines represent anion diffusion coefficient. Green strips indicate the bulk values ($2.26 \times 10^{-11}\,\mathrm{m^2\,s^{-1}}$ for the cations and $1.44 \times 10^{-11}\,\mathrm{m^2\,s^{-1}}$ for the anions, respectively) with the corresponding error bars.

the density profile is not affected by the metal polarisation effects. Further analysis of the first adsorbed layer, as provided by the density maps presented in Figure 2, reveals that the cations adsorb onto the surface, leaving vacant spots to be occupied by the anions. This lateral organisation shows a defective hexagonal arrangement of the anions. A similar ions distribution has been reported for [BMIM][PF_6] on graphite[34] where the lattice spacing in the adsorbed ions was about 4 times that of carbons in the graphite surface. The metal polarisation does not cause any significant change in the cation orientation in the first adsorbed layer (see Supplementary Information). Our picture of flat-lying imidazolium ions, separated by anions, which are only slightly above the cations plane is in agreement with experimental studies, where the orientation of imidazolium-based ionic liquids on Au(111) and Ni(111) has been probed by angle-resolved X-ray photoelectron spectroscopy (ARXPS),[35,36] on glass studied with X-ray photoelectron spectroscopy[37] and on Pt with Sum Frequency Generation.[38]

The organisation of the ions, creating variable local charge densities at the interface, in turn generate an electric field and hence a potential drop at the interface, according to Poisson's equation

$$\Phi = \Phi(\zeta) - \frac{1}{\varepsilon_0} \int_{\zeta}^{z} \mathrm{d}z' \int_{-\infty}^{z'} \mathrm{d}z'' \, \rho_q(z'') \quad (1)$$

where ζ is a reference in the left gold slab whose potential is $\Phi(\zeta)$ and $\rho_q(z)$ is the charge density. The potential profile shows an oscillating behaviour; a first positive peak corresponds to the closer adsorption of the cations, and it is followed by a negative peak, due to the anions layer. Due to their smaller size the anions can crowd in the second layer and overscreen the positive charge in the first layer. The fluctuation persist up to a few nanometers away from the gold surface. The potential of zero charge (PZC), defined here as the Poisson potential difference between the electrode and the bulk liquid, is 0.3 V and 0.26 V for non-polarisable and polarisable gold electrodes respectively. A similar value was found for an Au(100)/[BMIM][PF_6] system (PZC ≈ 0.48).[39] Overall, the gold polarisation at the interface seem to have a negligible effect. The addition of the image charges only slightly reduce the ion unbalance at the interface, therefore reducing the overall potential by only about 13 %.

At this point one may wonder why only such minor effects on the structure of the interfacial layer arise from the explicit introduction of the image charge effects. The reason may reside in the fact that locally the ion charges are screened by the surrounding counterions. Our analysis suggests that image charge effects are actually not so important as in vacuum, in analogy with the aqueous systems, which we have previously investigated[28], and where charges coming from solvated ions or from charged groups in solvated biomolecules are strongly screened by the water dielectric response. The static dielectric constant of [BMIM][BF_4], which is 13.9 at 298 K,[40] ($\sim 1/6$ of that of water), it is still large enough to reduce the electrostatic interactions by more than a factor of 10.

As next we would like to quantify and discuss the impact of the image charges on the local thermodynamic properties, namely on the interfacial free energy. We have calculated the work of adhesion, that is, the work needed to separate the liquid layer from the solid support, in both the non-polarisable and polarisable models- The Lennard-Jones contribution to the work of adhesion, reported in Figure 4 (A) as function of the coupling parameter, is qualitatively similar to that oibtained for other systems.[31,41,42] However, the work of adhesion for the ionic liquid, $W_{sl,LJ} = 485.9\,\mathrm{mJ\,m^{-2}}$, is much higher than that of water calculated on the same surface. This is not surprising, considering that RTILs have a much lower surface tension with respect e.g. to that of water, which corresponds to a higher wettability of the gold surface and to a stronger solid/liquid interaction. We now turn to the evaluation of the purely electrostatic contribution to the work of adhesion, which only results from the inclusion of the image charges We remind here that, at the PZC, in the absence of metal polarisation, there is no net charge on the gold and therefore no electrostatic interaction between the liquid and the surface. The Coulomb contribution to the work of adhesion $W_{sl,Coul} = 9.9\,\mathrm{mJ\,m^{-2}}$ is about 2 % of the total. This is at difference with the silica/water interface discussed by Surblys et al.,[42] where the Coulomb contribution accounts for about 15 % of the total work of adhesion, as result of to hydrogen bonds between water molecules and surface silanols.

In order to appreciate the impact of the image charge contribution to the interfacial free energy on the macroscopic properties, we calculate the change induced on the freezing temperature at the interface. According to the Gibbs-Thomson equation for the shift in freezing temperature temperature due to confinement, ΔT is given by[43]

$$\Delta T = T_{confined} - T_{bulk} = \frac{2 T_{bulk} \Delta\gamma}{\rho L_h \lambda_{width}} \quad (2)$$

where $T_{bulk} = 202\,\mathrm{K}$ is the freezing temperature of bulk [BMIM][BF_4], $\rho = 1211\,\mathrm{kg\,m^{-3}}$ its bulk density, $L_h = 4.69\,\mathrm{K\,kg^{-1}}$ its latent heat of fusion and $\Delta\gamma = \gamma_{lw} - \gamma_{sw}$ is the difference in the liquid-wall and solid-wall interfacial free energies. The γ_{lw} has been discussed already, while the γ_{sw} requires an additional calculation for the wall(gold)-solid(ionic liquid) interface. To

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this purpose we have generated two wall-solid interfaces freezing two independent configurations at temperature $T = 150\,\mathrm{K}$. For these two interfaces we have calculated, with the same dry surface approach the interface free energies, obtaining an average value of $\gamma_{sw} = -560.71\,\mathrm{mJ\,m^{-2}}$. Interestingly the image charge contribution for γ_{sw} is very close to that found for γ_{lw}. For a confinement width of $\lambda_{width} = 20\,\mathrm{nm}$, the freezing temperature raises by $\Delta T \approx 26.6\,\mathrm{K}$, which brings the bulk freezing temperature of $202.15\,\mathrm{K}$ to $228.75\,\mathrm{K}$, still quite far from room temperature. Clearly λ_{width} at the denominator implies that the freezing temperature may raise to room value as the confinement width is reduced to $5\,\mathrm{nm}$. Although the effect of confinement is quite important, the net contribution due to the image charge is actually negligible. This is essentially due to the fact that the solid [BMIM][BF_4] is actually amorphous, with a structure close to that of the interfacial liquid.

Having investigated, so far, the image charge effects on the structural as well as thermodynamical properties, a natural question arises on what is the impact of the image charges on transport and dynamical properties. We calculate here the translational self-diffusion of the two ionic components in the plane parallel to the surface (Figure 5) The size corrections[44] for the diffusion coefficients are, for the considered sizes, negligible (see S.I. for a detailed discussion). The bulk diffusion coefficient of the ions, calculated from five independent trajectories of 800 ionpairs at 298 K and including the size correction[45] is also reported, including the statistical uncertainty as a green stripe.

The cations diffuse much faster than the anions, with values monotonically increasing to the bulk value (Figure 5). From analysis of the structure, the interface only extends to about 2 nm from the gold surface. However, the dynamics shows that the interface extends about five times more, with the diffusion coefficient reaching bulk value only at 10 nm away from the gold surface. In particular the diffusion coefficient (Figure 5) reaches the bulk value according to an exponential law with a characteristic lengths λ between 2.9 nm and 4.5 nm (See TABLE **??**, Figure **??**). Interestingly this also implies an exponential dependence of the viscosity as function of the distance from the surface, which in turn would affect the viscous forces if non-equilibrium conditions are considered. Following the approach of Chan and Horn[46] and recently used by Perkin[26] to disentangle electrostatic and viscous contributions, the viscous contribution can be described with the Reynold formula:

$$F(t) = -\frac{6\pi\eta R^2 \dot{D}}{D-b} \tag{3}$$

where D is the distance between the solid surface in the SFB, η is the viscosity and b is the slip length. In their analysis of the experimental data the viscosity has been assumed as constant and a negative slip length has been introduced to take into account of the “frozen" interfacial layer.[26] From our simulations an interesting picture emerges where a “diffuse" layer may be defined not in terms of electrostatic, but instead of dynamical properties, as the region across which the liquid viscosity recovers its bulk value. Also for the self-diffusion of the ions, within the error bars, the non-polarisable and the polarisable gold models provide the same results. This last result suggests that viscous forces would be independent of the metallicity, at least under the investigated conditions, namely at the PZC.

3 Experimental section

Our system contains 1200 pairs of [BMIM][BF_4], confined between two gold slabs (electrodes) made each of 9 layers of gold atoms in the (111) crystallographic plane and spanning the $4.05457 \times 4.05457\,\mathrm{nm^2}$ cross-section of the box. Outside the electrodes a vacuum region of 222 nm is added. 2D periodic boundary conditions are enforced along the x and y directions, while the interaction between images along the z-directions are decoupled using the Particle Mesh Ewald with slab correction.[47] Two models for the gold were considered, a non-polarisable gold model[48] and a polarisable one,[28] which includes the induced image charges. To describe the ionic liquid, we use a non-polarisable model of [BMIM][BF_4] with force field parameters that reproduce experimental heat of vapourisation, diffusion coefficient, ionic conductivity and shear viscosity of [BMIM][BF_4].[49] In order to ensure a statistically relevant sampling five independent initial configurations were sampled from an equilibration run of 20 ns at a constant temperature of 298 K using a Nosé-Hoover thermostat, and at a constant pressure of 1 bar using a semi-isotropic Berendsen barostat. NVT production runs were performed on each of the five sampled configurations for 50 ns with a timestep of 0.5 fs and using a velocity rescale thermostat at 298 K, collecting system information every 0.5 ps. Figure 1 shows a typical simulation system.. Calculations of works of adhesion used the dry-surface approach (details in Supplementary Information).[31,42] All simulations were performed with version 5.1.4 of the GROMACS simulation package.[50]

4 Conclusions

We have reported here a systematic analysis of the impact of the image charges on the structural, thermodynamical and dynamical properties of [BMIM][BF_4] confined between two Au(111) gold electrodes, using atomistic molecular dynamics simulations. Cations are overscreened by the subsequent layer of anions, forming the double layer. Charge oscillations persist up to a couple of nanometers away from the surface, although a preferential orientation of the cations only affect the very first adsorbed layer. From the structural point of view, at the PZC, the interfacial region is only 2 nanometers thick. On the other hand, from the dynamical point of view, we notice that the ions dynamics at the interface is considerably slowed down and bulk diffusion is only recovered after 10 nm away from the gold surface.

Contrary to the somewhat intuitive expectations the image charges do not play any appreciable role on the structure and the dynamics. This is due to the strong electrostatic screening which takes place at the interface with the metal surface. Calculations of works of adhesion have shown that the impact of the classical image charge potential on the shift of the freezing temperature at the interface is negligible, which suggests that the cause of the increased capillary freezing observed in the work of Ref. 23 must be sought somewhere else. Enhanced freezing at the interface may be due to more complex effects, including the charging of the tip,

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and out of the equilibrium conditions due to the squeezing of the liquid between the substrate and the tip. Further complications may arise from the nanostructuration of the metal surface, due e.g. to defects and reconstruction. It has been shown that separation of ions in nanoporous media can locally enhance the electrostatic effects,[51] also including image charge effects. More work is therefore required in order to include these further aspects into the modelling and to bring the simulations closer to the complex experimental conditions.

Acknowledgments

We are grateful to Benjamin Rotenberg and Mathieu Salanne for useful discussions.

Parts of this research were conducted using the supercomputer Mogon and/or advisory services offered by Johannes Gutenberg University Mainz (hpc.uni-mainz.de).

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 674979-NANOTRANS and from the Deutsche Forschungsgemeinschaft under the TRR146 project.

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